

PROJECT PART 1

GitHub Repository:

<https://github.com/rortizca/CSE-464-2026-rortizca>

Links to each commit:

- Feature 1:
<https://github.com/rortizca/CSE-464-2026-rortizca/commit/a763feed8abd8b9465a557530427a57c95f1ff6>
- Feature 2:
<https://github.com/rortizca/CSE-464-2026-rortizca/commit/733f29749bcad89ce9190dd48985d6e6eec0d4e7>
- Feature 3:
<https://github.com/rortizca/CSE-464-2026-rortizca/commit/68575452cc7001cbade2887fd60031bc87df7bf4>
- Feature 4:
<https://github.com/rortizca/CSE-464-2026-rortizca/commit/a7c7dd2f4b69898b560d97810f9883d622c36302>
- Features Merge:
<https://github.com/rortizca/CSE-464-2026-rortizca/commit/93d99135f7f73af625de8eb3015fe5c229e8c13>

Instructions

The collection includes a dedicated Java Class object “Graph.java”, which contains all the API’s to create and modify a DOT graph file.

Each function operates according to the specification listed in the assignment description.

- 1) Start by creating a Graph object using the command below

```
Graph test_graph = new Graph();
```

- 2) Feature 1: You can parse a specified DOT file by utilizing the function parseGraph(String filepath)

```
test_graph.parseGraph("example.dot");
```

Input Graph:

```
digraph {
```

```
  A -> B
```

```

    B -> C
    C -> A
}

```

Expected Output:

```

Number of nodes: 3
Nodes:
A
C
B
Edges:
A -> B
C -> A
B -> C
Number of Edges: 3

```

You can also output the graph to a string using the toString() function

```
String output = test_graph.toString();
```

As well, you could output the graph to a specified filename using the outputGraph(String filepath) function

```
test_graph.outputGraph("./outputPath");
```

- 3) Feature 2: You can add any number Node to any existing Graph by utilizing either the addNode(String label) or addNodes(String[] labels) functions

```
test_graph.addNode("A");
```

Shared input:

```

digraph {
A -> B
}

```

Expected Output:

```

Number of nodes: 3
Nodes:
B
A
C
Edges:
A -> B
Number of Edges: 1

```

Or

```

String[] nodes = {"C", "D"};
test_graph.addNodes(nodes);

```

Expected Output:

```

Number of nodes: 4
Nodes:
B
A
D
C

```

```
Edges:
A -> B
Number of Edges: 1
```

- 4) Feature 3: You can add an edge between two distinct nodes using the function `addEdge(String srcLabel, String dstLabel)`

```
test_graph.addEdge("A", "B");
```

Input Graph:

```
digraph {
A
B
}
```

Expected Output:

```
Number of nodes: 2
Nodes:
B
A
Edges:
A -> B
Number of Edges: 1
```

- 5) Feature 4: You can output an imported Graph into a DOT file using `outputDOTGraph(String path)` or output that same Graph into a .png formatted file with a graphics output using `outputGraphics(String path, String format)`.

```
test_graph.outputDotGraph("./outputPath");
```

Or

```
test_graph.outputGraphics(outputPath, "png");
```

Input Graph:

```
digraph {
A
B
A -> B
}
```

Expected Output:

